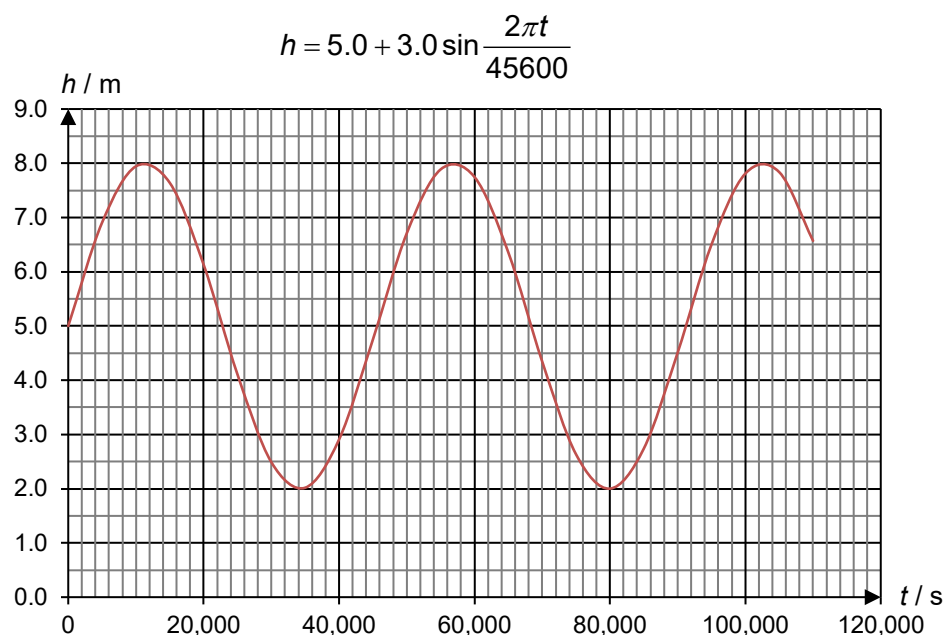


Tutorial 9 Oscillations Suggested Solutions

D1 (a)



- (i) h is maximum when $\sin \frac{2\pi t}{45600} = 1$
 $h = 5.0 + 3.0(1) = 8.0 \text{ m}$ (as can be seen from the graph above)
- (ii) h is minimum when $\sin \frac{2\pi t}{45600} = -1$
 $h = 5.0 + 3.0(-1) = 2.0 \text{ m}$ (as can be seen from the graph above)
- (iii) From the given equation,

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{45600}$$

$$T = 45600 \text{ s}$$
 time interval between high- and low-water is $\frac{1}{2}T = \frac{45600}{2} = 22800 \text{ s}$
- (iv) Water is 5.0 m deep every $\frac{1}{2}T$ i.e. $t = n\left(\frac{T}{2}\right)$ where $n = 0, 1, 2, 3, \dots$
 $t = 0 \text{ s}, 22800 \text{ s}, 45600 \text{ s}, 68400 \text{ s} \dots$
- (v) To find time when $h = 7.0 \text{ m}$,

$$7.0 = 5.0 + 3.0 \sin \frac{2\pi t}{45600}$$

$$\sin \frac{2\pi t}{45600} = \frac{7.0 - 5.0}{3.0}$$

$$t = \sin^{-1} \frac{2.0}{3.0} \times \frac{45600}{2\pi} = 5295.97 \text{ s}$$

Length of time for which h is more than 7.0 m,

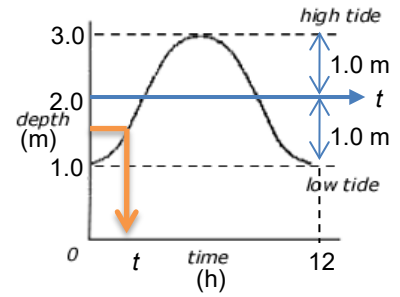
$$\frac{1}{2}T - 2(5295.97) = 12208 = 12200 \text{ s}$$

- (b) Since the tide change is simple harmonic, the 1.0 m (low tide) and 3.0 m (high tide) marks are taken as the extremes and the 2.0 m mark as the equilibrium point.

Thus the amplitude is 1.0 m and the S.H.M. equation is written as $x = -x_0 \cos \omega t = -(1.0 \text{ m}) \cos \omega t$. (boat enters at low tide)

The 1.5 m mark represents 0.5 m below the equilibrium point.

$$\begin{aligned} \text{Thus } -0.5 &= -(1.0) \cos \left\{ \left[\frac{2\pi}{12} \right] t' \right\} \\ \pi/3 &= \left[\frac{2\pi}{12} \right] t' \Rightarrow t' = 2.0 \text{ hours} \end{aligned}$$



Alternative Method:

Equation for the depth of water D in the harbour is:

$$D(t) = 2.0 - \cos \omega t$$

Waiting time for a boat at low tide for min depth of 1.5 m occurs when

$$1.5 = 2.0 - \cos \omega t$$

$$\cos \frac{2\pi}{T} t = 0.5 \Rightarrow t = \frac{\frac{\pi}{3}}{\frac{2\pi}{12}} = 2.0 \text{ hrs}$$

D2 (a) (i) $\theta = \omega t$

(ii) $ST = r \sin \theta = r \sin \omega t$

- (b)** Centripetal acceleration of peg = $\omega^2 r$

The acceleration a of the shadow on the screen is the component of the centripetal acceleration of the peg that is parallel to ST . Using the expressions in **(a)(i)** and **(a)(ii)**,

$$a = -\omega^2 r \sin \theta = -\omega^2 r \sin \omega t = -\omega^2 (ST) = -\omega^2 x$$

where x is the distance ST .

The negative sign indicates that the acceleration is directed towards S .

This gives us the defining equation of simple harmonic motion.

Hence the motion executed by the shadow is simple harmonic.

- (c) (i)** amplitude = radius of turntable = 20 cm

(ii) $T = \frac{2\pi}{\omega} = \frac{2\pi}{3.5} = 1.80 \text{ s}$

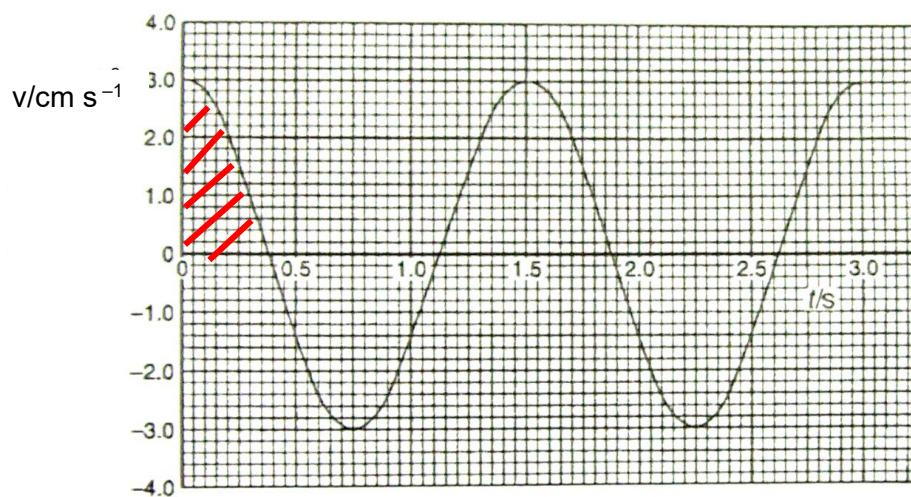
- (iii)** speed of shadow is maximum as it passes through the equilibrium position S ,
 $v_0 = \omega x_0 = (3.5)(0.20) = 0.700 \text{ m s}^{-1}$

- (iv)** magnitude of acceleration of the shadow is maximum when it is momentarily at rest,

$$a_o = \omega^2 x_o = (3.5)^2 (0.20) = 2.45 \text{ m s}^{-2}$$

- D3 (a)** Simple harmonic motion is defined as the motion of a particle about a fixed point such that its acceleration is proportional to its displacement from the fixed point and is always directed towards the point.

(b) (i)

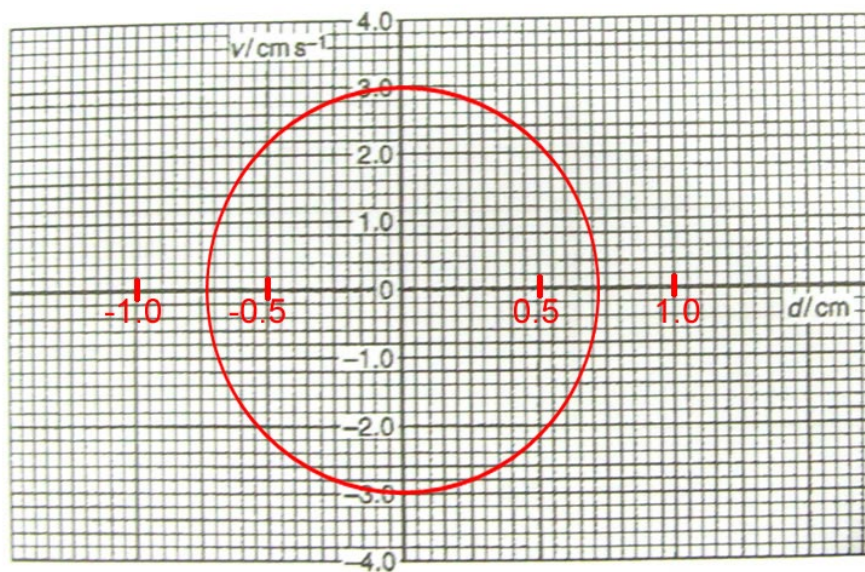


- (ii)**
1. From Fig. 2, period $T = 1.500 \text{ s}$ (read to half-the-smallest square)
Hence frequency $f = 1/T = 1/1.500 = 0.6667 \text{ Hz}$
 2. Amplitude = area of shaded area in **(b)(i)**
 $= 72 \text{ small squares} \times 2.0/10 \times 0.500/10$
 $= 0.72 \text{ cm}$

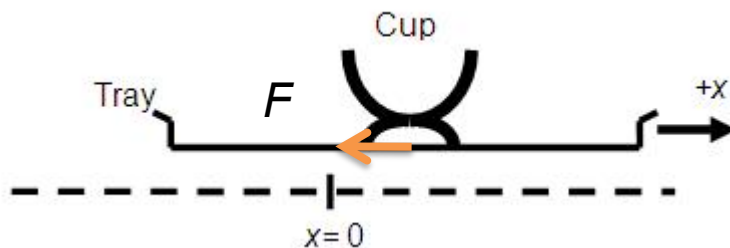
OR

$$\begin{aligned} \text{Apply } v_o &= \omega x_o \\ \Rightarrow 3.0 &= (2\pi/1.500) x_o \\ \Rightarrow x_o &= 0.72 \text{ cm} \end{aligned}$$

(iii)



D4 (a)



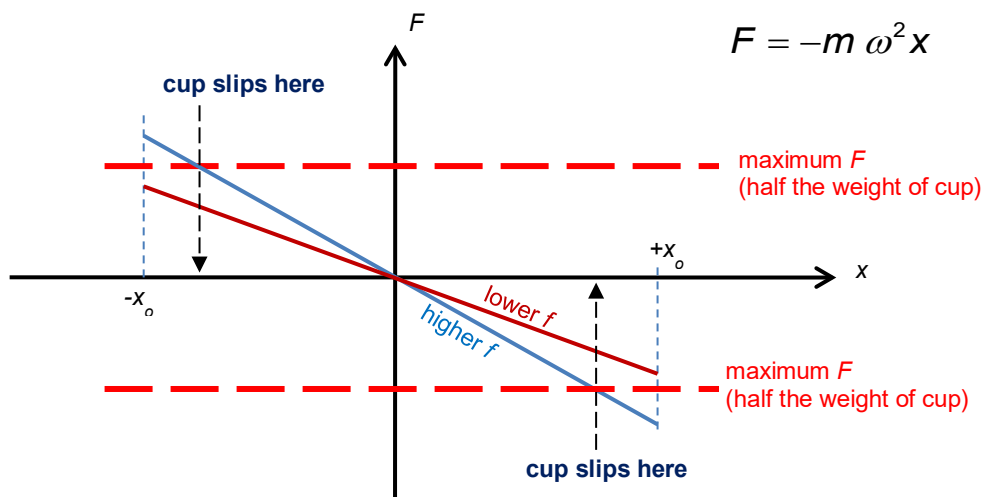
(b) Apply Newton's 2nd Law,

$$\therefore F_{\text{net}} = ma$$

$$\Rightarrow \text{friction} = m(-\omega^2 x)$$

$$\Rightarrow F = -m\omega^2 x$$

(c) The frictional force provides the restoring force for the cup to undergo S.H.M. The restoring force depends on ω and hence f . If the amplitude of oscillations x_0 is kept constant as f is increased, the restoring force required to maintain S.H.M. increases. When the required restoring force exceeds the maximum value of friction, the cup will not be brought back towards its equilibrium position. The cup will slip and stop undergoing S.H.M.



Note: $F = -m\omega^2 x = -(4\pi^2 mf^2)x$, i.e. higher frequency f means a steeper gradient.

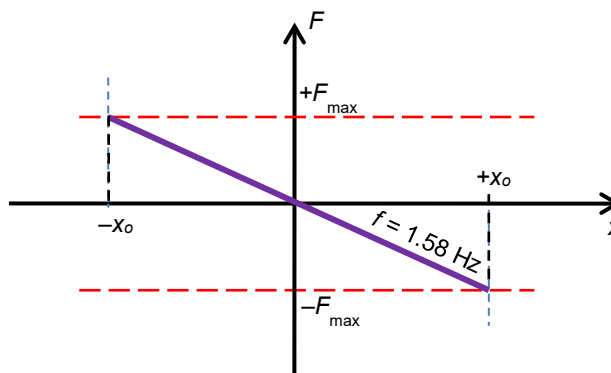
- (d) The maximum restoring force in S.H.M. occurs at the extremes of the oscillations because the restoring force is directly proportional to the displacement.

$$|F_{\max}| = m\omega^2 x_0$$

$$\frac{1}{2}mg = m\omega^2 x_0$$

$$\omega = \sqrt{\frac{9.81}{2(0.050)}} = 2\pi f$$

$$f = 1.576 = 1.58 \text{ Hz}$$



D5 Amplitude, $y_0 = 0.10 \text{ m}$

Period $T = 1.80 \text{ s}$

$$\omega = 2\pi/T = 3.49 \text{ rad s}^{-1}$$

- (a) maximum positive velocity:
- at $t = (n + \frac{3}{4})T$, where $n = 0, 1, 2, 3, \dots$ e.g. 1.35 s, 3.15 s, 4.95 s, etc.
 - at times where the gradient of $y-t$ graph is a positive maximum
- (b) maximum positive acceleration:
- at $t = (n + \frac{1}{2})T$, where $n = 0, 1, 2, 3, \dots$ e.g. 0.90 s, 2.70 s, 4.50 s, etc.
 - at times where y is negative maximum because $a-t$ and $y-t$ are in 'antiphase'
- (c) maximum negative acceleration:
- at $t = nT$, where $n = 0, 1, 2, 3, \dots$ e.g. 0.00 s, 1.80 s, 3.60 s, etc.
 - at times where y is positive maximum because $a-t$ and $y-t$ are in 'antiphase'
- (d) maximum kinetic energy:
- at times $t = (n/2 + \frac{1}{4})T$, where $n = 0, 1, 2, 3, \dots$
 - at times where the object has maximum positive and negative velocities, i.e. where the gradient of $y-t$ graph are positive maximum (1.35 s, 3.15 s, 4.95 s, etc.) and negative maximum (0.45 s, 2.25 s, 4.05 s, etc.)
- (e) maximum potential energy:
- at times $t = (n/2)T$, where $n = 0, 1, 2, 3, \dots$
 - at times where the object has maximum positive displacements (0.00 s, 1.80 s, 3.60 s, etc.) and negative maximum displacements (0.90 s, 2.70 s, 4.50 s, etc.)

D6 (a) (i) The straight line graph in the figure has a negative gradient and passes through the origin. Hence $a \propto -x$ which implies that the motion of P is simple harmonic as simple harmonic motion is defined as the motion of a particle about a fixed point such that its acceleration is proportional to its displacement from the fixed point and is always directed towards the point.

(ii) Using the coordinates: $(-0.20 \times 10^{-3}, 1700)$ and $(0.20 \times 10^{-3}, -1700)$,

$$\text{gradient} = -\omega^2 = \frac{1700 - (-1700)}{(-0.20 - 0.20) \times 10^{-3}} = \frac{3400}{-0.40 \times 10^{-3}}$$

$$\omega^2 = 8.5 \times 10^6$$

$$(2\pi f)^2 = 8.5 \times 10^6$$

$$f = \sqrt{\frac{8.5 \times 10^6}{4\pi^2}} = 464 = 460 \text{ Hz (2 s.f.)}$$

- (b) (i) The mass has maximum kinetic energy when it is at the equilibrium position ($x = 0$)

$$E_{\text{MAX}} = \frac{1}{2}mv_0^2 = \frac{1}{2}m(\omega A)^2 = \frac{1}{2}m\omega^2 A^2$$

$$\text{Hence, } E_{\text{MAX}} = \frac{1}{2}mGA^2, \text{ where } G = \omega^2$$

$$(ii) \quad E_{\text{MAX}} = \frac{1}{2}(2.5 \times 10^{-3})(8.5 \times 10^6)(0.4 \times 10^{-3})^2 = 1.7 \times 10^{-3} \text{ J}$$

- D7 (a)** From graph $T = 0.40 \text{ s}$

$$a_0 = \omega^2 x_0 = \left(\frac{2\pi}{T}\right)^2 x_0$$

$$= \left(\frac{2\pi}{0.40}\right)^2 (0.30 \times 10^{-2}) = 0.74 \text{ m s}^{-2}$$

$$\text{Maximum } E_p = \text{Total energy} = \frac{1}{2}m\omega^2 x_0^2$$

$$1.0 = \frac{1}{2}m\omega^2 x_0^2$$

$$m = \frac{2(1.0)}{\omega^2 x_0^2} = \frac{2.0}{a_0 x_0^2} = \frac{2.0}{(0.74)(0.30 \times 10^{-2})^2} = 900 \text{ kg}$$

- (b) (i) 1. $T = \frac{1}{f} = \frac{1}{835} = 1.198 \times 10^{-3} = 1.20 \times 10^{-3} \text{ s}$
2. The period of the energy-time graph is half the period of the molecule's vibration.
- $$(t_2 - t_1) = \frac{1}{2}T = \frac{1}{2}(1.198 \times 10^{-3}) = 5.988 \times 10^{-4} = 5.99 \times 10^{-4} \text{ s}$$
3. $v_{\text{max}} = \omega x_0 = (2\pi f) x_0 = 2\pi(835)(610 \times 10^{-9}) = 3.200 \times 10^{-3} = 3.20 \times 10^{-3} \text{ m s}^{-1}$
4. $E_{\text{max}} = \frac{1}{2}m(v_{\text{max}})^2 = \frac{1}{2}(5.3 \times 10^{-26})(3.200 \times 10^{-3})^2 = 2.714 \times 10^{-31} = 2.71 \times 10^{-31} \text{ J}$
- (ii) The speed of sound in a gas at room temperature is 340 m s^{-1} , which is more than 100 000 times the maximum speed of $3.20 \times 10^{-3} \text{ m s}^{-1}$ calculated in (i)3. of each gas molecule.

The speed of sound is the speed of propagation of energy by the sound wave, which does not involve the transfer of any medium. It is thus much greater than the maximum speed of the vibration of the medium (gas molecules) itself.

- D8 (a)** Since the tube is in equilibrium, the resultant force acting on the tube is zero.
Hence, the weight of the tube is equal and opposite to the upthrust acting on the tube.

*The upthrust is numerically equal to the weight of the fluid displaced by the floating tube.

$$mg = \rho Vg \quad \text{where } V \text{ is the volume of liquid displaced by the floating tube}$$

$$mg = \rho(Ah)g$$

$$m = \rho Ah$$

***You will not be expected to know that upthrust is numerically the weight of fluid displaced by the tube. However, you are expected to know that at equilibrium the 2 forces acting on the tube are the weight and upthrust, and that they are equal in magnitude and opposite in direction.**

- (b) (i)** Since ρ , A , g and m are constants, $a \propto -x$.

This implies that the acceleration a of the tube is directly proportional to the displacement x of the tube from its equilibrium position.

The negative sign implies that the acceleration a acts in the opposite direction to the displacement x . Since the direction of x is taken from the equilibrium position, the direction of a is towards the equilibrium position.

These satisfy the conditions for simple harmonic motion.

- (ii)** Compare $a = -\left(\frac{\rho Ag}{m}\right)x$ with the defining equation for SHM: $a = -\omega^2 x$

$$\omega^2 = \frac{\rho Ag}{m}$$

$$(2\pi f)^2 = \frac{\rho Ag}{m}$$

$$f = \sqrt{\frac{\rho Ag}{4\pi^2 m}} = \sqrt{\frac{(1.0 \times 10^3)(4.2 \times 10^{-4})(9.81)}{4\pi^2 (32 \times 10^{-3})}} = 1.806 = 1.8 \text{ Hz (shown)}$$

- (c) (i) 1.** From Fig. 8.2,
 $T = 0.50 \text{ s}$
 $f = \frac{1}{T} = \frac{1}{0.50} = 2.0 \text{ Hz}$

2.

$$\omega^2 = \frac{\rho Ag}{m}$$

$$(2\pi f)^2 = \frac{\rho Ag}{m}$$

$$\rho = \frac{4\pi^2 f^2 m}{Ag} = \frac{4\pi^2 (2.0)^2 (32 \times 10^{-3})}{(4.2 \times 10^{-4})(9.81)} = 1226 = 1.23 \times 10^3 \text{ kg m}^{-3}$$

- (ii) 1. As the tube oscillates, there is work done against air resistance and the viscous force due to the fluid. This causes the oscillating system to lose energy continuously to the surroundings.

Since there is no continuous supply of energy to the system, the total energy of the oscillating system will decrease with time.

The decrease in total energy is exhibited in the decrease in the amplitude x_0 of the oscillation as total energy E_T is given by $\frac{1}{2}m(v_{\max})^2 = \frac{1}{2}m\omega^2 x_0^2$.

2. At $t = 0.00 \text{ s}$, $x_0 = 1.50 \times 10^{-2} \text{ m}$
At $t = 1.00 \text{ s}$, $x_0 = 0.85 \times 10^{-2} \text{ m}$

decrease in total energy

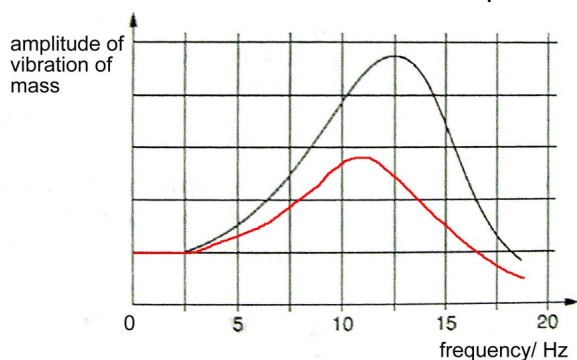
$$\begin{aligned} &= \frac{1}{2}m\omega^2 (1.50 \times 10^{-2})^2 - \frac{1}{2}m\omega^2 (0.85 \times 10^{-2})^2 \\ &= \frac{1}{2}(32 \times 10^{-3})(2\pi \times 2.0)^2 \left[(1.50 \times 10^{-2})^2 - (0.85 \times 10^{-2})^2 \right] \\ &= 3.859 \times 10^{-4} \\ &= 3.86 \times 10^{-4} \text{ J} \end{aligned}$$

D9 (a) Resonance

(b) (i) $\omega = 2\pi f = 2\pi(12.5) = 78.5 \text{ rad s}^{-1}$

(ii) $T = \frac{1}{f} = \frac{1}{12.5} = 0.0800 \text{ s}$

- (c) The light piece of card increases the damping of the oscillating mass. This will decrease the maximum amplitude at resonance and the resonance frequency



- (d) A radio tuning circuit, magnetic resonance imaging.

D10 (a) Resonance occurs when a system responds at maximum amplitude to an external driving force. This occurs when the frequency of the driving force is equal to the natural frequency of the driven system, where there is maximum transfer of energy to the driven system.

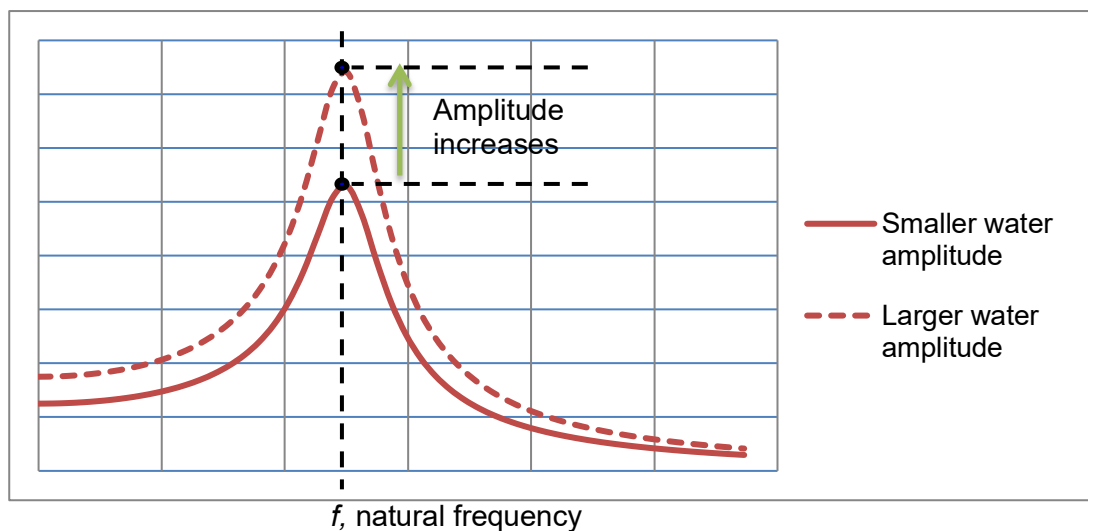
(b) (i) $f = \frac{v}{\lambda} = \frac{0.90}{0.30} = 3.0 \text{ Hz}$

(ii) At resonance,
driving frequency of water waves = natural frequency:

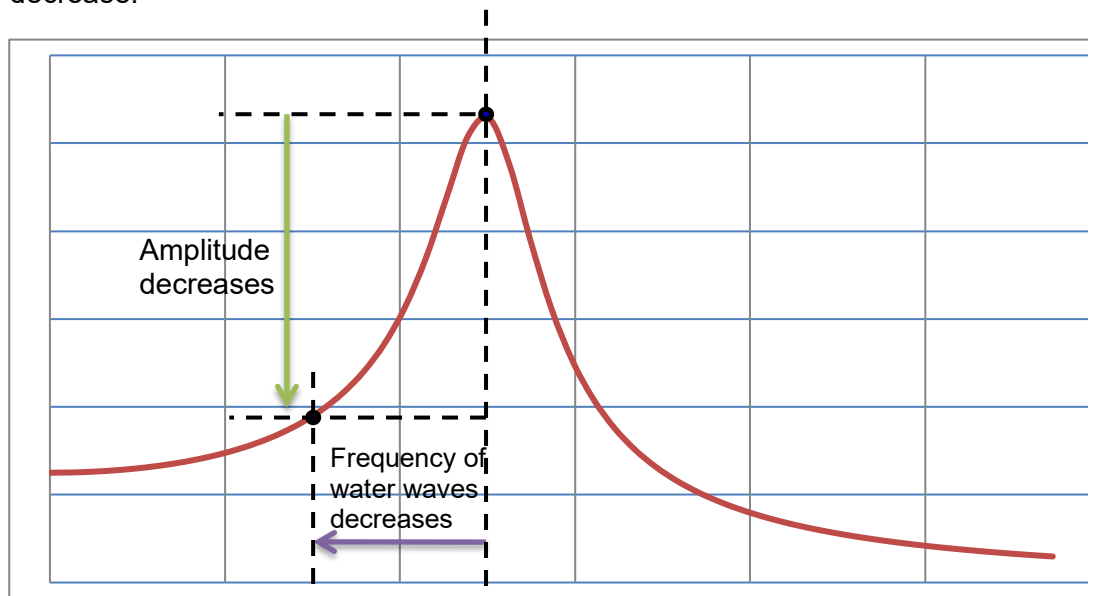
$$3.0 = \frac{1}{2\pi} \sqrt{\frac{28}{m}}$$

$$m = \left(\frac{1}{6\pi} \right)^2 (28) = 0.079 \text{ kg}$$

(c) (i) The amplitude of the oscillations is larger at resonance because of the increased rate of transfer of energy from the water waves of larger amplitude. ($P \propto f^2 A^2 v$)

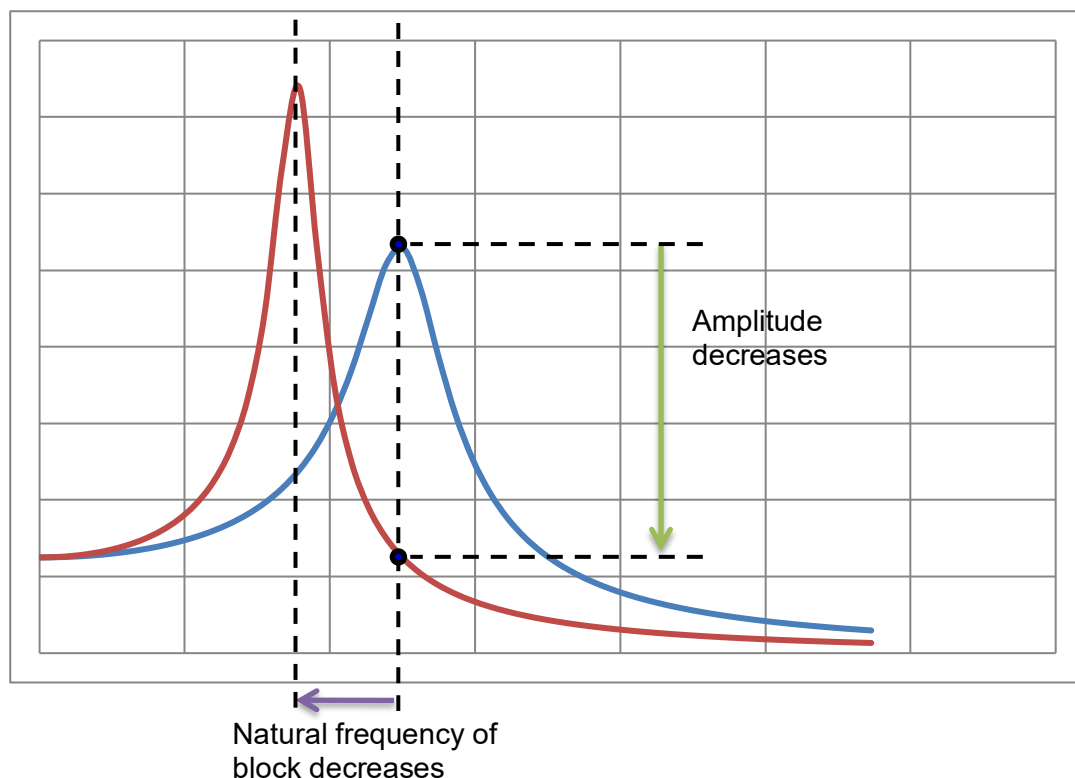


(ii) When the wavelength of the water waves increases, its frequency will decrease (and become smaller than the block's natural frequency f) and there will no longer be resonance. Hence, the amplitude of the vertical oscillations of the block will decrease.



(iii) The natural frequency of the block will be reduced (and become smaller than the driving frequency) as a result of the increase in its mass and it is no longer at resonance. Hence, at the same driver frequency, the amplitude of the vertical

oscillations of the block will decrease. (The peak of the curve is shifted to the left, but it need not be higher than the original curve.)



- C1 (a)** Since the centre of mass of the system (one point P on the spring) is stationary, the oscillation of m_1 is then effectively due to a spring of natural length l_1 (where l_1 is the distance of P from the equilibrium position of m_1) and appropriate force constant k_1 , fixed at P. Similarly, the oscillation of m_2 is effectively due to a spring of natural length l_2 (where l_2 is the distance of P from the equilibrium position of m_2) and appropriate force constant k_2 , fixed at P.

Assume m_1 and m_2 oscillate with different amplitudes A_1 and A_2 respectively. Since the centre of mass is stationary, we have

$$m_1 A_1 = m_2 A_2 \quad \text{———— (1)}$$

As the spring exerts equal and opposite forces on both masses,

$$F = k_1 A_1 = k_2 A_2$$

Equation (1) becomes

$$m_1 A_1 = m_2 A_2 \Rightarrow m_1 \frac{F}{k_1} = m_2 \frac{F}{k_2} \Rightarrow \frac{m_1}{k_1} = \frac{m_2}{k_2} \Rightarrow T_1 = T_2$$

- (b) Let m_1 moves through a distance x to the left and m_2 a distance y to the right.
Centre of mass of system is stationary:

$$m_1 x = m_2 y$$

$$y = \frac{m_1}{m_2} x$$

Change in length of the spring

$$= x + y = x + \left(\frac{m_1}{m_2} x \right) = x \left(1 + \frac{m_1}{m_2} \right)$$

- (c) Assume k to be the force constant of the spring.

If the extension of the spring is e , then acceleration of m_1 is given by

$$\begin{aligned} a_1 &= -\frac{k}{m_1} e \\ &= -\frac{k}{m_1} \left[x \left(1 + \frac{m_1}{m_2} \right) \right] \\ &= -k \left(\frac{m_1 + m_2}{m_1 m_2} \right) x \end{aligned}$$

Hence,

$$\begin{aligned} \omega^2 &= k \left(\frac{m_1 + m_2}{m_1 m_2} \right) \\ \frac{2\pi}{T} &= \sqrt{k \left(\frac{m_1 + m_2}{m_1 m_2} \right)} \\ T &= 2\pi \sqrt{\frac{m_1 m_2}{k(m_1 + m_2)}} \end{aligned}$$

- C2** Mass m_2 shoots off, carrying E_k away from the system. To find its speed v :
Elastic PE of spring = maximum KE of masses

$$\begin{aligned} \frac{1}{2} k x^2 &= \frac{1}{2} (m_1 + m_2) v^2 \\ v^2 &= \frac{k x^2}{(m_1 + m_2)} \end{aligned}$$

With m_1 alone on the spring, the energy balance equation becomes

$$\text{maximum } E_p = \text{maximum } E_k$$

$$\frac{1}{2} kx_o^2 = \frac{1}{2} m_1 v^2$$

$$= \frac{1}{2} m_1 \left(\frac{kx^2}{(m_1 + m_2)} \right)$$

$$x_o^2 = x^2 \left(\frac{m_1}{m_1 + m_2} \right)$$

$$x_o = x \sqrt{\frac{m_1}{m_1 + m_2}}$$